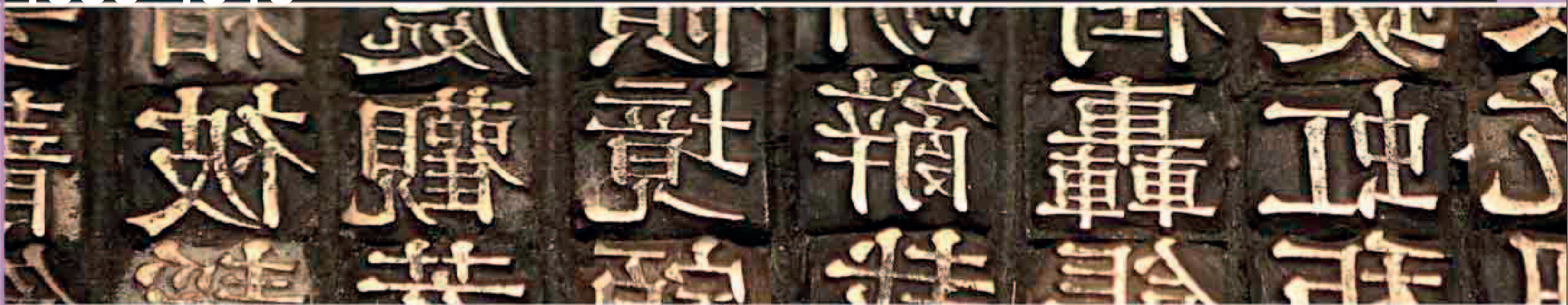
1030–1049



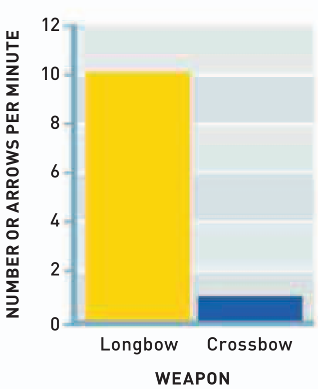
The earliest movable type in China was made of clay, and later of wood. Metal movable type such as these blocks did not become common until the Ming dynasty in the 17th century.

IN THE EARLY 11TH CENTURY, Spanish–Arab astronomer Abu **Abdallah ibn Mu'adh al-Jayyani** (989–1079) carried out work integrating trigonometry and optics. His *Book of Unknown Arcs of a Sphere* was the first comprehensive work on spherical trigonometry. Around 1030, al-Jayyani used this work in his *Book on Twilight* to calculate the angle of the Sun below the horizon at the end of evening twilight to be 18 degrees. By taking this as the lowest angle at which the Sun's rays can meet the upper edge of the atmosphere, he worked out the height of Earth's atmosphere as 64 miles (103 km).

Printing using carved wooden blocks had appeared in China around the 6th century, but the process was cumbersome, requiring a new block to be carved for each individual page. Around 1040, a commoner named **Pi Sheng** developed **a form of movable type** by creating thin strips of clay, each impressed with a single character, which he baked in a fire. He then placed these on an iron tray to compose the page to be printed. The clay letters could be rearranged as desired to create a new page. The method fell into disuse after Pi Sheng's death until its revival in the mid-13th century. By then, far **more durable type made of iron** had been invented in Korea, where it was **first used in 1234**.

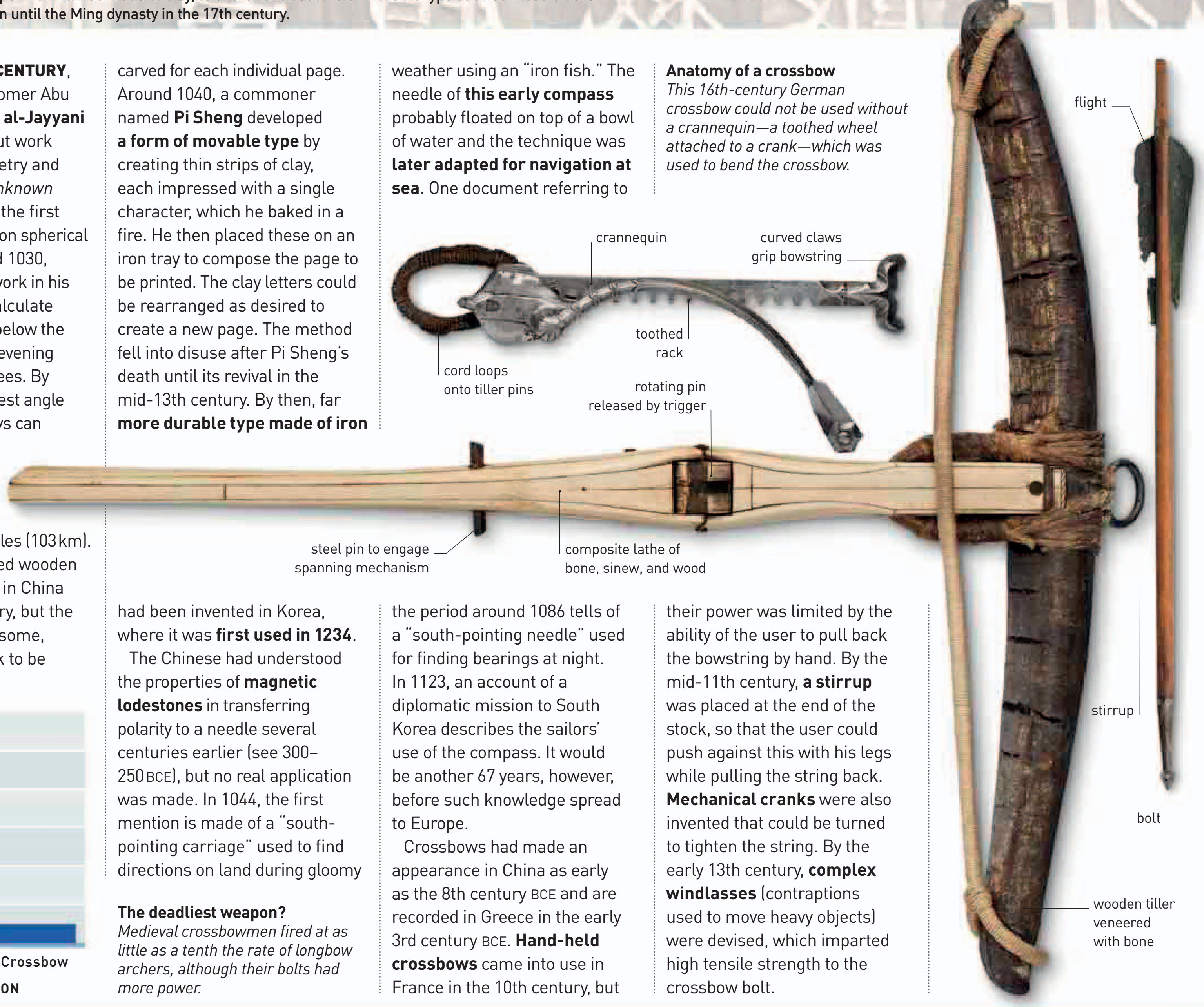
The Chinese had understood the properties of **magnetic lodestones** in transferring polarity to a needle several centuries earlier (see 300–250 BCE), but no real application was made. In 1044, the first mention is made of a "south-pointing carriage" used to find directions on land during gloomy weather using an "iron fish." The needle of **this early compass** probably floated on top of a bowl of water and the technique was **later adapted for navigation at sea**. One document referring to the period around 1086 tells of a "south-pointing needle" used for finding bearings at night. In 1123, an account of a diplomatic mission to South Korea describes the sailors' use of the compass. It would be another 67 years, however, before such knowledge spread to Europe.

Crossbows had made an appearance in China as early as the 8th century BCE and are recorded in Greece in the early 3rd century BCE. **Hand-held crossbows** came into use in France in the 10th century, but their power was limited by the ability of the user to pull back the bowstring by hand. By the mid-11th century, **a stirrup** was placed at the end of the stock, so that the user could push against this with his legs while pulling the string back. **Mechanical cranks** were also invented that could be turned to tighten the string. By the early 13th century, **complex windlasses** (contraptions used to move heavy objects) were devised, which imparted high tensile strength to the crossbow bolt.



The deadliest weapon?
Medieval crossbowmen fired at as little as a tenth the rate of longbow archers, although their bolts had more power.

Anatomy of a crossbow
This 16th-century German crossbow could not be used without a crannequin—a toothed wheel attached to a crank—which was used to bend the crossbow.



c.1030 Persian astronomer Al-Biruni suggests that **Earth may rotate around the Sun**, but says he cannot prove it

c.1030 Guido d'Arezzo devises **new theory of musical notation** and develops system of hexachords

c.1030 Al-Nasawi summarizes Euclid's *Elements* and describes method for extraction of **cube roots**

c.1040 Pi Sheng invents **movable type for printing** using clay blocks

1044 First mention of **magnetic compass used for navigation**, in China

c.1050 Ibn Butlan writes *Taqwim al-sihha* treatise emphasizing **importance of good diet and hygiene**